

Theory of Automata

Assignment - 1

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Section: BSCS - ~~1E~~ 3E

Section 1

Q1a) All string of length greater than or equal to 3

RegEx: $(a+b)^*(a+b)(a+b)(a+b)(a+b)^*$

Rough combo $\Rightarrow$

ϵ	a	a	a	ϵ
ϵ	b	b	b	b
aa	b	a	b	bb
bbb	b	b	a	aaa

Q1b) All strings with atleast 2a's in them

RegEx: $(a+b)^*a(a+b)^*a(a+b)^*$

Rough combo $\Rightarrow$

ϵ	a	ϵ	a	ϵ
a	a	a	a	a
aa	a	bbb	a	aaa
bbb	a	bb	a	b

Q1c) All strings with at most 2a's in them

RegEx: $b^* + (b^*ab^*) + b^*ab^*ab^*$

Rough combo:

ϵ a is 0	a is 1 max	a is 2 max
bbbb	ϵ a ϵ	
	b a b	
	bb a bb	
	bbb a ϵ	
		ϵ a ϵ a ϵ $\approx$ aa
		b a b a b
		bb a b a ϵ

Q1d) All strings with atleast two occurrences of bb in them

RegEx: $(a+b)^* bb (a+b)^* bb (a+b)^*$

rough combo $\Rightarrow \epsilon \quad bb \quad \epsilon \quad bb \quad \epsilon$

aa bb aa bb aa

bbb bb bbbb bb bbbbbb

a bb bbb bb bb

Q1e) All strings containing even number of b's

RegEx: $a^* + (a^* ba^* ba^*)^*$

rough combo $\Rightarrow$

$\downarrow$ b is 0 so it's even
 ϵ
 a
 aaa
 aaaaa

$\downarrow$
 $\epsilon b \epsilon b \epsilon$
abaabaa abaabaa
 $\epsilon baaab \epsilon$ $\epsilon baaab \epsilon$ $\epsilon baaab \epsilon$

$(a+b)^*$
 $\uparrow$
 $(a+b)(a+b)$
 $\uparrow$
 $(a+b)(a+b)(a+b)$

Q1f) All strings not containing bab

RegEx: $(a + bb^*aa)^* (\epsilon + bb^* + bb^*a)$

$\underbrace{\hspace{1cm}}$ ensures never bab
 $\underbrace{\hspace{1cm}}$ allows for string to end at a
 $\underbrace{\hspace{1cm}}$ strings end at b
 $\underbrace{\hspace{1cm}}$ end at a

rough combos:

a

ϵ

aaa

ϵ

bbaa

ϵ

bbaa

bbbb

bbbbaa

bbbba

aaaa

bbbbbb

aa aa

bbba

Q1g) All strings that start and end with different letters
eg (ab, ba, baa, aab)

$$\text{RegEx : } \underbrace{(a(a+b)^*b)}_{\substack{\swarrow \\ \text{gives} \\ a \text{ (string) } b}} + \underbrace{(b(a+b)^*a)}_{\substack{\swarrow \\ \text{gives} \\ b \text{ (string) } a}}$$

Q1h) All strings where each instance of aa must be followed by 'b'. eg (aab, aabab, aababab)

$$\text{RegEx : } (b+ab+aab)^*(\epsilon+a)$$

rough combo $\Rightarrow$

ϵ	ϵ	aab	a
ϵ	a	aabab	a
b	ϵ	aababab	ϵ
b	a	aabababab	a
ab	ϵ		
ab	a		

Q1i) All strings which contain aa or bb or both

$$\text{RegEx : } \underbrace{(a+b)^*}_{\substack{\swarrow \\ \text{can give either a} \\ \text{combo of "aa"} \\ \text{or "bb"}}} \underbrace{(aa+bb)}_{\substack{\searrow \\ \text{ensures atleast} \\ \text{either "aa" or} \\ \text{"bb" exists}}} \underbrace{(a+b)^*}$$

rough combo $\Rightarrow$

ϵ	aa	$\epsilon\epsilon$
ϵ	bb	$\epsilon\epsilon$
a	bb	bbb
aa	aa	aa
bb	aa	bbabbb

Q1j) All strings with an even length but an odd number of a's

RegEx: $(aa+bb)^*(ab+ba)(aa+bb)^*$

rough combo $\Rightarrow$

ϵ	ab	ϵ	even length odd a
ϵ	ba	ϵ	
aa	ba	bb	even length add a
$\underline{aa} \underline{bb} \underline{aa}$	ab	$\underline{bb} \underline{bb}$	even length odd a

Q2a) Requirement:

- start with exactly 2 digits
- followed by a word $\rightarrow$ (can be a-z, A-Z, 0-9)
- than a literal "+" symbol
- followed by another word (can be a-z, A-Z, 0-9)
- than @MyEmailService.com
- No space, dot, underscore, or special character

RegEx: $^{[0-9]\{2\}[A-Za-z0-9]+\backslash+[A-Za-z0-9]+@MyEmailService\.$
com \$

Breakdown of above RegEx

$\wedge \rightarrow$ start of string

$[0-9]\{2\} \rightarrow$ exactly two digits

$[A-Za-z0-9]^+ \rightarrow$ first word (can comprise of multiple small/capital letters and/or digits)

$\backslash+ \rightarrow$ a literal "+" symbol present in string

$[A-Za-z0-9]^+ \rightarrow$ second word (one or more letters or digits)

$@MyEmailService\.$ com $\rightarrow$ must end with exactly this domain
 $\rightarrow$ literal "." symbol

\$ $\rightarrow$ end of string

Q2b) Requirement :

- Must start with capital english alphabet
- Length ≥ 14
- Must contain at least one digit and at least one special character.

RegEx : $^((?=.{14,})\$)(?=.*\d)(?=.*[^\wdd])([A-Z]\S{13})\$$

Breakdown of above regular expression

$^$ $\rightarrow$ start of string

$(?=.{14,})\$$ $\rightarrow$ a lookahead ensuring the entire string is at least 14 characters long

$\downarrow$
14 or more repetition
 $\downarrow$ match any character.
 $\downarrow$ look ahead syntax

$(?=.*\d)$ $\rightarrow$ a lookahead ensuring the entire string has at least one digit

$\downarrow$
 $\downarrow$ a digit (0-9)
 $\downarrow$ zero or more of any characters that can be before this ~~doesn't~~
 $\downarrow$ lookahead syntax

$(?=.*[^\wdd])$ $\rightarrow$ a lookahead ensuring the entire string has at least one special character

$\downarrow$
 $\downarrow$ a character that is NOT uppercase A-Z, lowercase a-z, or digit (0-9)
 $\downarrow$ zero or more of any characters
 $\downarrow$ look ahead syntax

$[A-Z]$ $\rightarrow$ The first character must be uppercase A-Z

$\S{13}$ $\rightarrow$ after the first uppercase letter, there must be at least 13 non-whitespace characters.

$\downarrow$
 $\downarrow$ non-whitespace character

$\$$ $\rightarrow$ end of string

Q2c) Requirements

- Must contain either "comp" or "imp"
- Must have a 'z' followed by at least one "o"
- Must end with "virus"

RegEx: $^((?=.*(comp|imp)))(?=.*zo+).+virus\$$

Breakdown of above regular expression

$\wedge$ → start of string

$(?=.*(comp|imp))$ → a lookahead ensuring either "comp" or "imp" must exist somewhere in the string.

→ a GROUP with alternation: matches either "comp" or "imp"

→ zero or more of any characters

→ lookahead syntax

$(?=.*zo+)$ → a lookahead ensuring string contains "z" followed by at least one or more "o".

→ literal "z"

→ zero or more of any character

→ lookahead syntax

→ one or more "o" eg 0, oo, ooo ----

$.+virus$ → string must end with virus

→ literal substring "virus"

→ zero or more of any character

$\$$ → end of string

Section 2

Q1a) What is the initial state?

M1: q_1

M2: q_1

Q1b) What is the set of acceptance states?

M1: $\{q_2\}$

M2: $\{q_1, q_4\}$

Q1c) What sequence of states does the machine go through for the following inputs:

i) aabb

M1: q_1, q_2, q_3, q_1, q_1

sequence of flow

M2: q_1, q_1, q_2, q_4

ii) bbaa

M1: q_1, q_1, q_2, q_3

M2: q_1, q_2, q_4, q_3, q_2

Q1d) Does the machine accept the input aabb?

M1: No, it does not

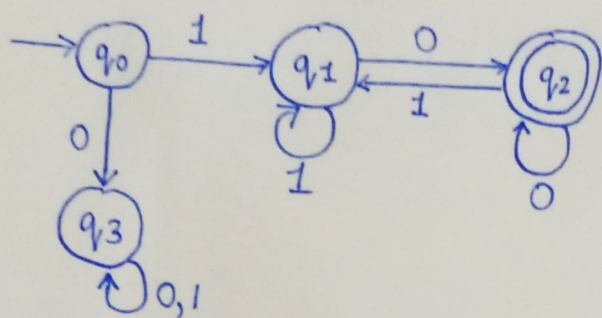
M2: Yes, it does

Q1e) Does the machine accept the input ϵ ?

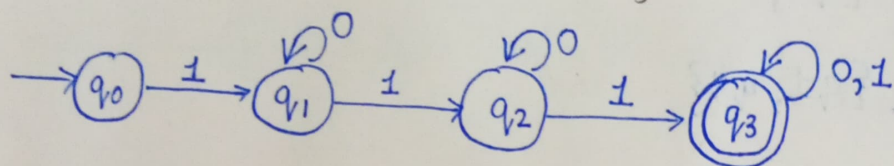
M1: No, it does not

M2: Yes, it does

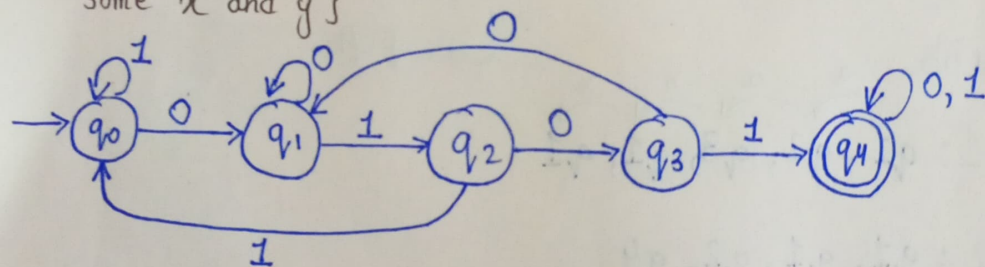
Q2a) $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$



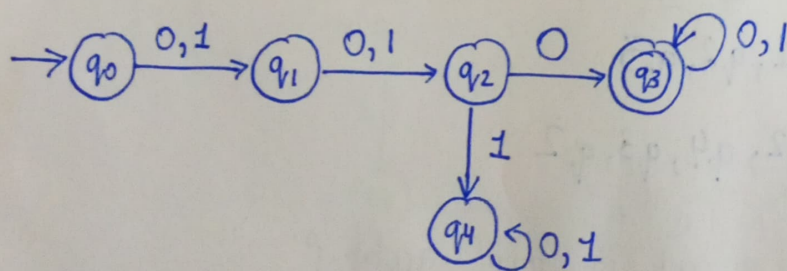
Q2b) $\{w \mid w \text{ contains at least three 1's}\}$



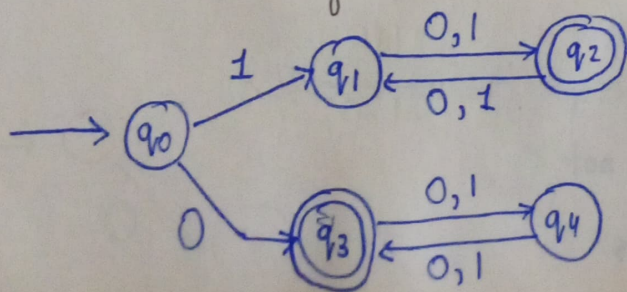
Q2c) $\{w \mid w \text{ contains the substring 0101, i.e., } w = x0101y \text{ for some } x \text{ and } y\}$



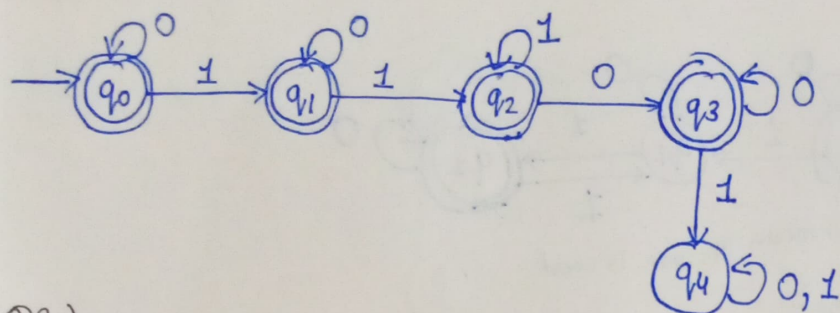
Q2d) $\{w \mid w \text{ has length of at least 3 and its third symbol is a 0}\}$



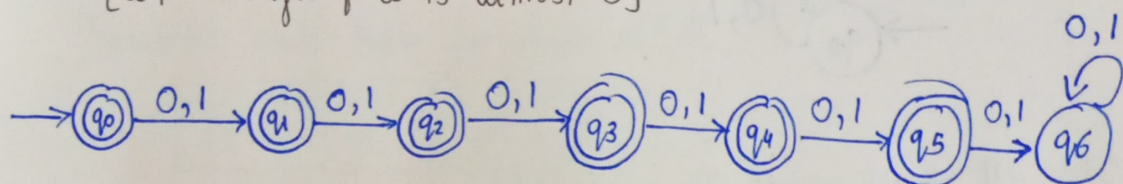
Q2e) $\{w \mid w \text{ starts with 0 and has odd length or starts with 1 and has even length}\}$



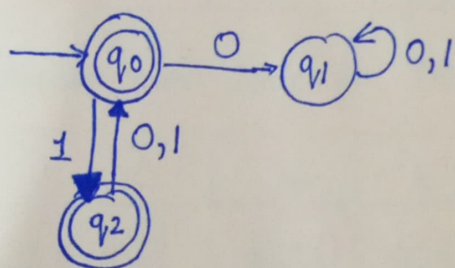
Q2f) $\{w \mid w \text{ doesn't contain the substring } 1101\}$



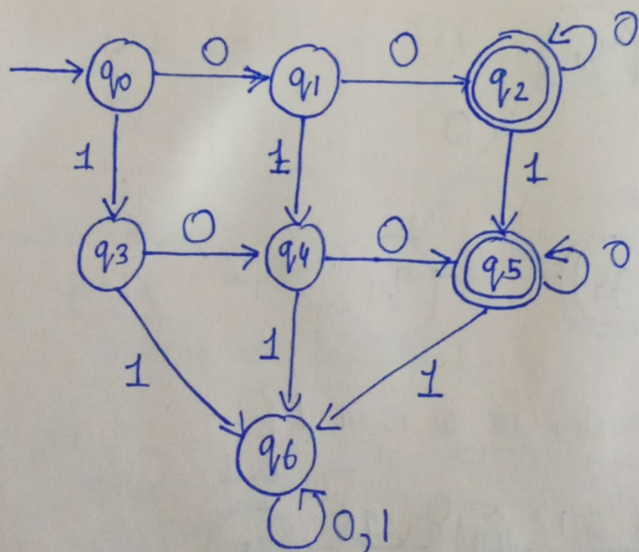
Q2g) $\{w \mid \text{the length of } w \text{ is at most } 5\}$



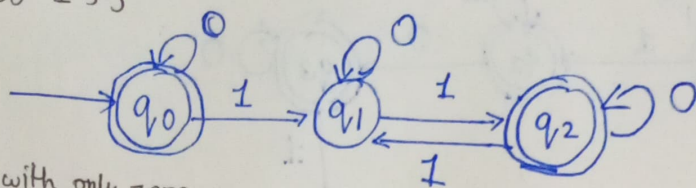
Q2h) $\{w \mid \text{every odd position of } w \text{ is a } 1\}$



Q2i) $\{w \mid w \text{ contains at least two } 0\text{'s and at most one } 1\}$

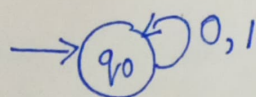


Q2j) $\{w \mid w \text{ contains an even number of 1's, or contains exactly two 1's}\}$

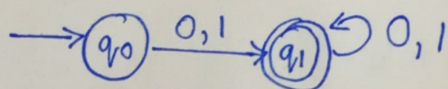


* a string with only zero means the one is even

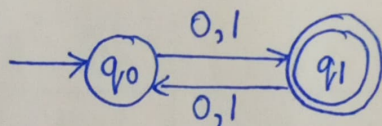
Q2k) The empty set



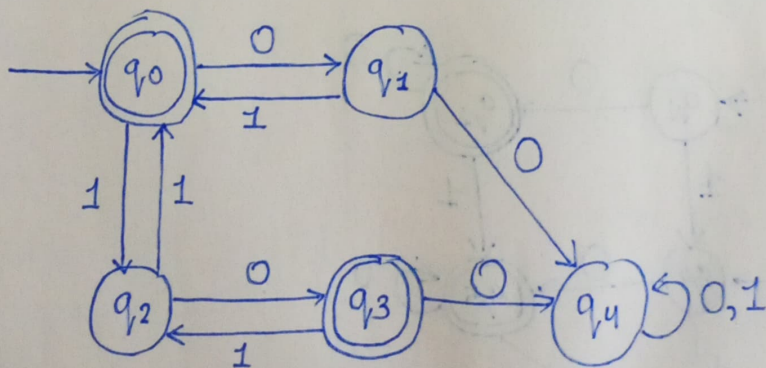
Q2l) All strings except the empty string



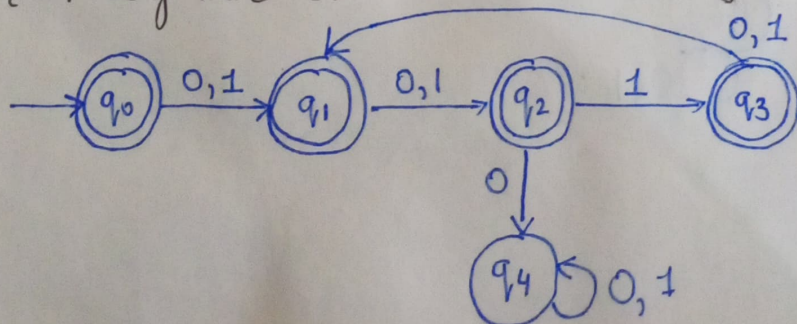
Q2m) $\{w \mid \text{length of } w \text{ is odd}\}$



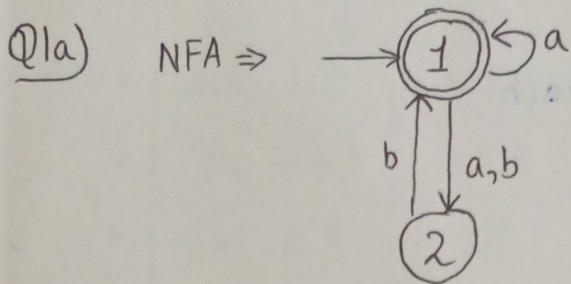
Q2n) $\{w \mid \text{length of } w \text{ is even and it must not contain } 00\}$



Q2o) $\{w \mid \text{every third character in } w \text{ is a 1}\}$



Section 3



Sol:

Transition state table for above NFA

δ	a	b
$\rightarrow^*$ 1	{1, 2}	{2}
2	∅ -	{1, 2} {1}

Transition state table for DFA corresponding to above NFA

δ	a	b
Let A = {1}	{1, 2}	{2}
B = {1, 2}	{1, 2}	{1, 2}
C = {2}	D	{1}
D	D	D

working
 when state A = {1}
 for a
 $1 \rightarrow 1$
 $1 \rightarrow 2 \therefore A \xrightarrow{a} \{1, 2\}$
 for b
 $1 \rightarrow 2 \therefore A \xrightarrow{b} \{2\}$

• when state C = {2}
 for a
 $2 \rightarrow$ no move (dead state)

for b
 $2 \rightarrow 1$
 $\therefore C \xrightarrow{b} \{1\}$

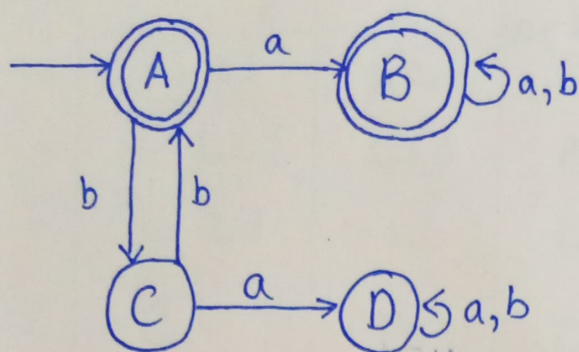
• Dead State
 denoted by "D"

• when state B = {1, 2}
 for a
 $1 \rightarrow 1$ and $2 \rightarrow 1$
 $1 \rightarrow 2$

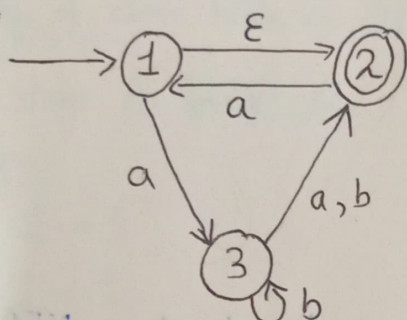
$\therefore \{1, 2\} \cup \{1\}$
 $\therefore B \xrightarrow{a} \{1, 2\}$

for b
 $1 \rightarrow 2$
 $2 \rightarrow 1$
 $\therefore B \xrightarrow{b} \{1, 2\}$

DFA State Diagram



Q1b) NFA



Sol

Transition table for above NFA

	δ	ϵ	a	b
$\rightarrow$ 1		{2}	{3}	-
* 2		-	{1}	-
3		-	{2}	{2, 3}

Epsilon (?) - closure

? closure {1} : {1,2}

? closure {2} : {2}

? closure {3} : {3}

two epsilon

no epsilon

no epsilon

Converting NFA with ϵ -move (epsilon closure move) to DFA

δ	a	a	b
$\rightarrow * \text{ Let } A =$	1, 2	1, 2, 3	D
* B =	1, 2, 3	1, 2, 3	2, 3
* C =	2, 3	1, 2	2, 3
	D	D	D

working of above table of DFA

- when state $A = \{1, 2\}$

for a

$$1 \rightarrow 3$$

$$2 \rightarrow 1$$

$$\text{move}(A, a) = \{1, 3\}$$

$$\epsilon\text{-closure}\{1, 3\} = \epsilon\text{-cl}\{1\} \cup \epsilon\text{-cl}\{3\}$$

$$= \{1, 2\} \cup \{3\}$$

$$= \{1, 2, 3\}$$

$$\therefore A \xrightarrow{a} \{1, 2, 3\}$$

for b

$$1 \rightarrow \text{no move}$$

$$2 \rightarrow \text{no move}$$

$$\therefore A \xrightarrow{b} \text{Dead State}$$

- when state $B = \{1, 2, 3\}$

for a

$$1 \rightarrow 3$$

$$2 \rightarrow 1$$

$$3 \rightarrow 2$$

$$\text{move}(B, a) = \{3, 1, 2\}$$

$$\epsilon\text{-closure}\{3, 1, 2\} = \epsilon\text{-cl}\{3\} \cup \epsilon\text{-cl}\{1\} \cup \epsilon\text{-cl}\{2\}$$

$$= \{3\} \cup \{1, 2\} \cup \{2\}$$

$$= \{1, 2, 3\}$$

$$\therefore B \xrightarrow{a} \{1, 2, 3\}$$

for b

$$1 \rightarrow \text{none}$$

$$2 \rightarrow \text{none}$$

$$3 \rightarrow 2, 3$$

$$\text{move}(B, b) = \{2, 3\}$$

$$\epsilon\text{-cl}\{2, 3\} = \epsilon\text{-cl}\{2\}$$

$\cup$

$$\epsilon\text{-cl}\{3\}$$

$$= \{2\} \cup \{3\}$$

$$= \{2, 3\}$$

$$B \xrightarrow{b} \{2, 3\}$$

• when state $C = \{2, 3\}$

for a

$2 \rightarrow 1$

$3 \rightarrow 2$

$\text{move}(C, a) = \{1, 2\}$

$$\begin{aligned} \epsilon\text{-closure}(\{1, 2\}) &= \epsilon\text{-cl}\{1\} \cup \epsilon\text{-cl}\{2\} \\ &= \{1, 2\} \cup \{2\} \\ &= \{1, 2\} \end{aligned}$$

$\therefore C \xrightarrow{a} \{1, 2\}$

for b

$2 \rightarrow \text{no move}$

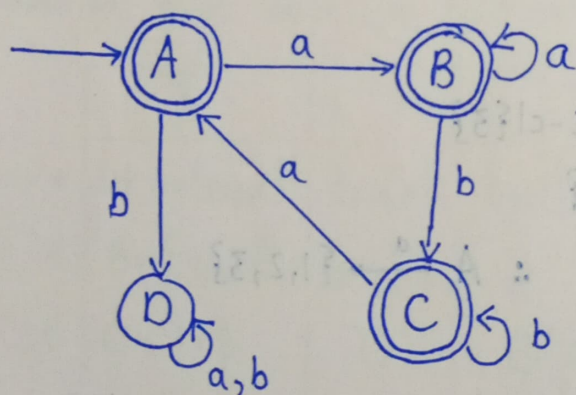
$3 \rightarrow 3, 2$

$\text{move}(C, b) = \{2, 3\}$

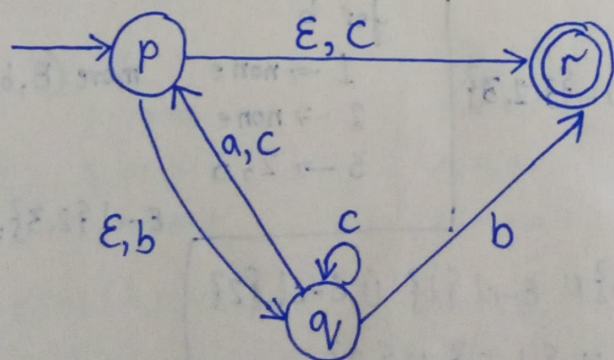
$$\begin{aligned} \epsilon\text{-closure}(\{2, 3\}) &= \epsilon\text{-cl}\{2\} \cup \\ &\quad \cup \epsilon\text{-cl}\{3\} \\ &= \{2\} \cup \{3\} \\ &= \{2, 3\} \end{aligned}$$

$\therefore C \xrightarrow{b} \{2, 3\}$

DFA state diagram



Q1c) ϵ -NFA state Diagram



Sol:

Transition state table for above ϵ -NFA diagram

	δ	ϵ	a	b	c
$\rightarrow$	p	$\{q, r\}$	-	$\{q\}$	$\{r\}$
	q	-	$\{p\}$	$\{r\}$	$\{p, q\}$
*	r	-	-	-	-

Epsilon ? closure :

$$\epsilon\text{-closure}(p) : \{p, q, r\}$$

$$\epsilon\text{-closure}(q) : \{q\}$$

$$\epsilon\text{-closure}(r) : \{r\}$$

Transition state table for DFA corresponding to above NFA

	δ	a	b	c
$\rightarrow$ *	Let $A = \{p, q, r\}$	$\{p, q, r\}$	$\{q, r\}$	$\{p, q, r\}$
*	$B = \{q, r\}$	$\{p, q, r\}$	$\{r\}$	$\{p, q, r\}$
*	$C = \{r\}$	$\emptyset$	$\emptyset$	$\emptyset$
	D	$\emptyset$	$\emptyset$	$\emptyset$

working of above table

• when state $A = \{p, q, r\}$

for a

$$p \rightarrow \text{none}$$

$$q \rightarrow p$$

$$r \rightarrow \text{none}$$

$$\text{move}(A, a) = \{p\}$$

$$\epsilon\text{-cl}(p) = \epsilon\text{-cl}(\{p\})$$

$$= \{p, q, r\}$$

for b

$$p \rightarrow q$$

$$q \rightarrow r$$

$$r \rightarrow \text{none}$$

$$\text{move}(A, b) = \{q, r\}$$

$$\epsilon\text{-cl}(\{q, r\}) = \epsilon\text{-cl}(\{q\}) \cup \epsilon\text{-cl}(\{r\}) = \{q, r\}$$

for c

$$p \rightarrow r$$

$$q \rightarrow q, p$$

$$r \rightarrow \text{none}$$

$$\text{move}(A, c) = \{p, q, r\}$$

$$\epsilon\text{-cl} = \epsilon\text{-cl}(\{p\}) \cup \epsilon\text{-cl}(\{q\}) \cup \epsilon\text{-cl}(\{r\})$$

$$= \{p, q, r\}$$

- when state $B = \{q, r\}$

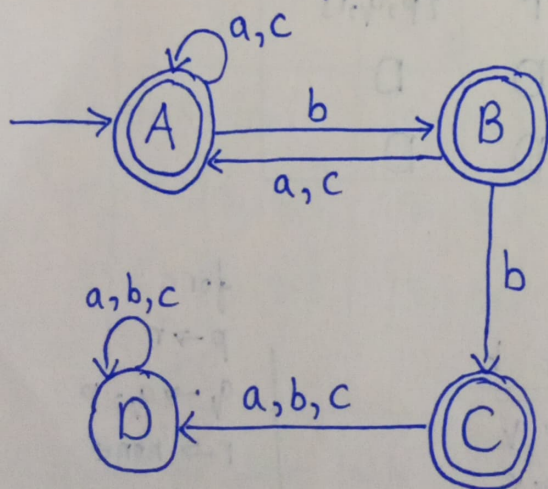
<u>for a</u>	<u>for b</u>	<u>for c</u>
$q \rightarrow p$	$q \rightarrow r$	$q \rightarrow q, p$
$r \rightarrow \text{none}$	$r \rightarrow \text{none}$	$r \rightarrow \text{none}$
$\text{move}(B, a) = \{p\}$	$\text{move}(B, b) = \{r\}$	$\text{move}(B, c) = \{p, q\}$
$\epsilon\text{-cl } \{p\} = \{p, q, r\}$	$\epsilon\text{-cl } \{r\} = \{r\}$	$\epsilon\text{-cl } \{p, q\} = \epsilon\{p\} \cup \epsilon\text{-cl } \{q\}$
$\therefore B \xrightarrow{a} \{p, q, r\}$	$\therefore B \xrightarrow{b} \{r\}$	$= \{p, q, r\} \cup \{q\}$
		$= \{p, q, r\}$

$$\therefore B \xrightarrow{c} \{p, q, r\}$$

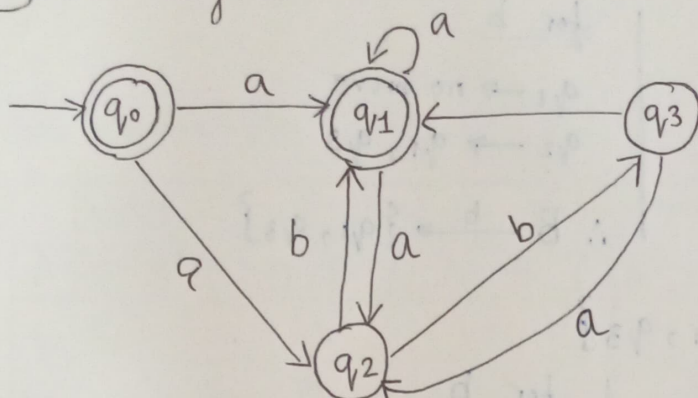
- when state $C = \{r\}$

<u>for a</u>	<u>for b</u>	<u>for c</u>
$r \rightarrow \text{none}$	$r \rightarrow \text{none}$	$r \rightarrow \text{none}$
Dead state	Dead state	dead state

DFA State Diagram



Q1d) NFA diagram



Sol:

Transition state table for above NFA

δ	a	b
$\rightarrow \star q_0$	$\{q_1, q_2\}$	-
$\star q_1$	$\{q_1, q_2\}$	-
q_2	-	$\{q_1, q_3\}$
q_3	$\{q_1, q_2\}$	-

Transition state table for DFA corresponding to above NFA

δ	a	b
$\rightarrow \star \text{Let } A = \{q_0\}$	$\{q_1, q_2\}$	D
$\star B = \{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_1, q_3\}$
$\star C = \{q_1, q_3\}$	$\{q_1, q_2\}$	D
D	D	D

working for above table

• when state $A = q_0$

for a

$q_0 \rightarrow q_1, q_2$

$\therefore A \xrightarrow{a} \{q_1, q_2\}$

for b

$q_0 \rightarrow \text{no move}$

$\therefore A \xrightarrow{b} \text{dead state}$

• when state $B = \{q_1, q_2\}$

for a

$q_1 \rightarrow q_1, q_2$

$\therefore B \xrightarrow{a} \{q_1, q_2\}$

for b

$q_1 \rightarrow \text{no move}$

$q_2 \rightarrow q_1, q_3$

$\therefore B \xrightarrow{b} \{q_1, q_3\}$

• when state $C = \{q_1, q_3\}$

for a

$q_1 \rightarrow q_1, q_2$

$\therefore C \xrightarrow{a} \{q_1, q_2\}$

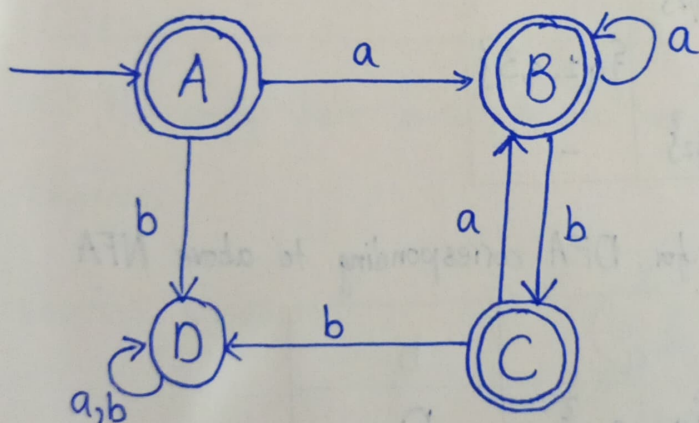
for b

$q_1 \rightarrow \text{no move}$

$q_3 \rightarrow \text{no move}$

$\therefore C \xrightarrow{b} \text{Dead state}$

DFA State Diagram



Section 4

Q1

state transition table

S	states										output									
	A	B	C	D	E	F	G	H	I		A	B	C	D	E	F	G	H	I	
S ₁	S ₂										0									
S ₂		S ₃	S ₉									0	1							
S ₃				S ₆	S ₄									0	0					
S ₄				S ₆	S ₅									0	0					
S ₅				S ₆	S ₉									0	2					
S ₆						S ₇										0				
S ₇							S ₈										0			
S ₈							S₈	S ₆	S ₉									0	1	
S ₉																				

All the empty space in output section are 0 indicating nothing happened.

Transition Diagram:

